FR-03 – Step 3: Select test cases (best representative of each class)

Input: the classes in `FR-03_2_equivalence_classes.md`. Method: lecture S04, step 3: - Choose **at least one** test case from each equivalence class. - **Valid** classes: each test case covers as many valid classes as possible. - **Invalid** classes: each test case covers **one and only one** invalid class. All its other inputs come from valid classes, so a failure points to exactly one cause.

Columns follow the course template `Test cases_Template_En.xlsx`. Actual result, status, tester and date are filled in at step 5 (execution).

Common setup

ID	Setup
P1	Backend (`:3000`) and frontend-web (`:5173`) running. Database freshly seeded with `node database.js` , so `test@eshop.com` / `Test1234!` exists.
P2	A second account, **user B**, registered through the Register page: `userb@eshop.com` / `UserB123!` .
P3	**Step 2 ready:** open `http://localhost:5173/forgot-password` , enter `test@eshop.com` , submit. The page shows step 2 and an OTP; call it **X**.
API	API test cases use Postman or curl. `POST http://localhost:3000/api/forgot-password` returns the OTP in `resetToken` (API spec 1.3). `POST http://localhost:3000/api/reset-password` takes `email` , `resetToken` , `newPassword` (API spec 1.4).
Cleanup	After any test case that changes the password (expected or not), re-seed the database and redo P2. This keeps every test case independent of the others.

Default valid values used whenever a test case doesn't target that field: email `test@eshop.com`, OTP = **X**, new password `NewPass1!` , confirm `NewPass1!` .

Valid test cases

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-03_TC_01	Reset password with all-valid inputs (happy path)	EC_01, 03, 07, 13, 15, 18, 21, 24, 26, 29, 31, 33, 35, 40, 45, 47, 50, 54, 60; outputs EC_62, 63, 64, 74, 75, 77	P1	1. Open forgot-password page; check the step indicator. 2. Email `test@eshop.com` , submit. 3. Note OTP X; check the step indicator. 4. OTP `X` , new password `NewPass1!` , confirm `NewPass1!` , submit within 1 minute. 5. Log in with `NewPass1!` . 6. Log out, log in with `Test1234!` .	1. "Bước 1 / 2" shown. 3. 6-digit OTP shown; "Bước 2 / 2" shown. 4. Success message; redirected to login. 5. Login succeeds. 6. Login fails.

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-03_TC_02	"Back to login" on step 1	EC_11; output EC_76	P1	1. Open forgot-password page. 2. Click "Back to login".	Login page is shown.
FR-03_TC_03	"Back to login" on step 2	EC_12; output EC_76	P3	1. On step 2, click "Back to login".	Login page is shown.
FR-03_TC_04	Use the latest of two OTPs; new password equals current	EC_55, EC_39	P1	1. Request an OTP for `test@eshop.com` (OTP X1). 2. Request again (OTP X2). 3. OTP `X2`, new password `Test1234!`, confirm `Test1234!`, submit.	Success message; login with `Test1234!` works.
FR-03_TC_05	OTP typed with surrounding spaces	EC_20	P3	1. OTP `" X "` (one space before and after), defaults for the rest, submit.	Success (spaces are trimmed).
FR-03_TC_06	Correct OTP that starts with 0	EC_23	P1	1. Repeat step 1 until the OTP shown starts with `0` (about 1 try in 10). 2. Enter that OTP exactly, defaults for the rest, submit.	Success.
FR-03_TC_07	Reset API with a matching confirm field	EC_57	P1	1. Get OTP X via API. 2. `POST /api/reset-password {"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!","confirmPassword":"NewPass1!"}`	HTTP 200; password changed.

Invalid test cases – step 1 (email)

Precondition for all: P1. Each submits only the email shown.

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_08	Empty email	EC_02; output EC_67	Email `""` (also try `"  "`), submit.	Required-field error; stays on step 1; no OTP.
FR-03_TC_09	Email without `@`	EC_04; output EC_66	Email `testeshop.com`, submit.	Invalid-format error; no OTP.
FR-03_TC_10	Email with empty local part	EC_05	Email `@eshop.com`, submit.	Invalid-format error; no OTP.
FR-03_TC_11	Email without top-level domain	EC_06	Email `test@eshop`, submit.	Invalid-format error; no OTP.
FR-03_TC_12	Well-formed but unregistered email	EC_08; output EC_65	Email `nobody@eshop.com`, submit.	"Not registered" error; no OTP.
FR-03_TC_13	Registered email in different case	EC_09	Email `Test@eshop.com`, submit.	"Not registered" error (tester decision).

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_14	Registered email with surrounding spaces	EC_10	Email `" test@eshop.com "`, submit.	Invalid-format error (tester decision).

Invalid test cases – step 2 (OTP)

Precondition for all: P3. New password and confirm are `NewPass1!` unless stated.

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_15	Empty OTP	EC_14	OTP `""`, submit.	Required-field error.
FR-03_TC_16	OTP shorter than 6	EC_16; output EC_68	OTP = first 5 digits of X, submit.	Incorrect-OTP error.
FR-03_TC_17	OTP longer than 6	EC_17	OTP = X followed by `1` (7 digits), submit.	Incorrect-OTP error.
FR-03_TC_18	OTP containing a letter	EC_19	OTP = X with its 3rd digit replaced by `a`, submit.	Incorrect-OTP error.
FR-03_TC_19	Wrong 6-digit OTP; password unchanged	EC_22; output EC_78	1. OTP = X with its last digit changed (e.g. +1, 9→0), submit. 2. Log in with `Test1234!`.	1. Incorrect-OTP error. 2. Login succeeds (password unchanged).
FR-03_TC_20	Expired OTP	EC_46; output EC_69	Wait **15 minutes** after getting X, then submit with OTP X. (T is not in the spec; 15 min is assumed to be past any reasonable expiry.)	OTP-expired error.
FR-03_TC_21	Reuse an OTP after a successful reset	EC_48; output EC_70	1. Submit OTP X with `NewPass1!` (succeeds). 2. Request nothing new; open step 2 again (or use the API) with OTP X, password `Other123!`.	2. OTP-already-used error; `Other123!` does not log in.
FR-03_TC_22	Reuse an OTP after a failed reset	EC_49	1. Submit OTP X with password `short 1!` (fails, too weak). 2. Submit OTP X again with `NewPass 1!`.	2. OTP-already-used error (tester decision).
FR-03_TC_23	Older OTP after a newer one was requested	EC_56	1. Request OTP X1, then request OTP X2. 2. Submit OTP `X1`.	Incorrect-OTP error.

Invalid test cases – step 2 (new password)

Precondition for all: P3. OTP = X; confirm = the same value as the new password, so only the password rule is wrong.

TC ID	Test case name	Class covered	New password and confirm	Expected result
FR-03_TC_24	Empty password	EC_25	`""`	Required-field error.
FR-03_TC_25	Too short (7)	EC_27; output EC_72	`Pass12!`	Weak-password error.

TC ID	Test case name	Class covered	New password and confirm	Expected result
FR-03_TC_26	Very long (129)	EC_28	`Aa1!` + 125 × `a`	Weak-password / length error (tester decision); no crash.
FR-03_TC_27	No uppercase	EC_30	`newpass1!`	Weak-password error.
FR-03_TC_28	No lowercase	EC_32	`NEWPASS1!`	Weak-password error.
FR-03_TC_29	No digit	EC_34	`NewPass!!`	Weak-password error.
FR-03_TC_30	No special character	EC_36	`NewPass12`	Weak-password error.
FR-03_TC_31	Special character only outside the set	EC_37	`NewPass1#`	Weak-password error.
FR-03_TC_32	Contains a space	EC_38	`New Pass1!`	Weak-password error (tester decision).

Invalid test cases – step 2 (confirm password)

Precondition for all: P3. OTP = X; new password = `NewPass1!`.

TC ID	Test case name	Class covered	Confirm value	Expected result
FR-03_TC_33	Empty confirm	EC_41	`""`	Required-field error.
FR-03_TC_34	Completely different	EC_42; output EC_73	`Other123!`	Passwords-don't-match error.
FR-03_TC_35	Differs only in case	EC_43	`newpass1!`	Passwords-don't-match error.
FR-03_TC_36	Differs only by a trailing space	EC_44	`"NewPass1! "`	Passwords-don't-match error.

Invalid test cases – reset API

Precondition for all: P1 + P2. Get OTP X for `test@eshop.com` via `POST /api/forgot-password` unless stated. Each request uses valid values except the one field named.

TC ID	Test case name	Class covered	Request body for `POST /api/reset-password`	Expected result
FR-03_TC_37	OTP of user A used for user B	EC_51; output EC_71	Also request an OTP for user B. Then `{"email":"userb@eshop.com","resetToken":"X","newPassword":"NewPass1!"}`	Error "OTP not valid for this email"; user B's password unchanged.
FR-03_TC_38	Unregistered email	EC_52	`{"email":"nobody@eshop.com","resetToken":"X","newPassword":"NewPass1!"}`	Error; no account changed.
FR-03_TC_39	Email missing	EC_53	`{"resetToken":"X","newPassword":"NewPass1!"}`	Error; no account changed.
FR-03_TC_40	Confirm field present but not matching	EC_58	`{"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!","confirmPassword":"Other123!"}`	Passwords-don't-match error; password unchanged.

TC ID	Test case name	Class covered	Request body for `POST /api/reset-password`	Expected result
FR-03_TC_41	Confirm field absent	EC_59 (TBD)	`{"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!"}` (exactly API spec 1.4)	**Decided at execution.** Record what happens; the spec conflict is reported either way.
FR-03_TC_42	Reset for an email that never requested an OTP	EC_61	Do **not** request an OTP for user B. `{"email":"userb@eshop.com","resetToken":"123456","newPassword":"NewPass1!"}`	Error; user B's password unchanged.

GUI test cases – FR-21 / FR-22 (added after review)

These check how each step looks and behaves, not its input domains. Each covers the GUI output classes on one page.

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-03_TC_43	Step 1 page meets the form and UI rules	Outputs EC_79, 80, 82, 83, 84, 85, 86	P1	1. Open the forgot-password page. 2. Inspect the email field (right-click → Inspect). 3. Count `<h1>` elements (DevTools: `document.querySelectorAll('h1').length`). 4. Press Tab repeatedly from the top of the page. 5. Submit with the email empty.	1. All text in Vietnamese; submit button blue; `*` next to the email label. 2. `type="email"`. 3. Exactly 1. 4. Focus moves top to bottom, left to right. 5. Error message appears **above** the submit button, in Vietnamese.
FR-03_TC_44	Step 2 page meets the form and UI rules	Outputs EC_79, 81, 82, 83, 84, 85, 86	P3	1. On step 2, check labels and button. 2. Type in the new password and confirm fields; inspect them. 3. Count `<h1>` elements. 4. Press Tab repeatedly from the top. 5. Submit with OTP X, new password `abc` and confirm `abc`.	1. Vietnamese text; blue submit button; `*` next to OTP, new password and confirm labels. 2. Characters masked; both `type="password"`. 3. Exactly 1. 4. Focus order top to bottom, left to right. 5. Error appears **above** the submit button, in Vietnamese.

Coverage check

Every class from step 2 is covered by at least one test case, and each invalid test case targets exactly one invalid input class.

Class	Test case		Class	Test case		Class	Test case
EC_01	TC_01		EC_27	TC_25		EC_53	TC_39
EC_02	TC_08		EC_28	TC_26		EC_54	TC_01

Class	Test case		Class	Test case		Class	Test case
EC_03	TC_01		EC_29	TC_01		EC_55	TC_04
EC_04	TC_09		EC_30	TC_27		EC_56	TC_23
EC_05	TC_10		EC_31	TC_01		EC_57	TC_07
EC_06	TC_11		EC_32	TC_28		EC_58	TC_40
EC_07	TC_01		EC_33	TC_01		EC_59	TC_41
EC_08	TC_12		EC_34	TC_29		EC_60	TC_01
EC_09	TC_13		EC_35	TC_01		EC_61	TC_42
EC_10	TC_14		EC_36	TC_30		EC_62	TC_01
EC_11	TC_02		EC_37	TC_31		EC_63	TC_01
EC_12	TC_03		EC_38	TC_32		EC_64	TC_01
EC_13	TC_01		EC_39	TC_04		EC_65	TC_12
EC_14	TC_15		EC_40	TC_01		EC_66	TC_09
EC_15	TC_01		EC_41	TC_33		EC_67	TC_08
EC_16	TC_16		EC_42	TC_34		EC_68	TC_16
EC_17	TC_17		EC_43	TC_35		EC_69	TC_20
EC_18	TC_01		EC_44	TC_36		EC_70	TC_21
EC_19	TC_18		EC_45	TC_01		EC_71	TC_37
EC_20	TC_05		EC_46	TC_20		EC_72	TC_25
EC_21	TC_01		EC_47	TC_01		EC_73	TC_34
EC_22	TC_19		EC_48	TC_21		EC_74	TC_01
EC_23	TC_06		EC_49	TC_22		EC_75	TC_01
EC_24	TC_01		EC_50	TC_01		EC_76	TC_02
EC_25	TC_24		EC_51	TC_37		EC_77	TC_01
EC_26	TC_01		EC_52	TC_38		EC_78	TC_19
						EC_79	TC_43, TC_44
						EC_80	TC_43
						EC_81	TC_44
						EC_82	TC_43, TC_44
						EC_83	TC_43, TC_44
						EC_84	TC_43, TC_44
						EC_85	TC_43, TC_44
						EC_86	TC_43, TC_44

Summary

	Count
Valid test cases	7
Invalid test cases	34

	Count
Test case with expected result decided at execution	1 (TC_41)
GUI test cases (FR-21/FR-22, added after review)	2
Total test cases	**44**
Classes covered	86 / 86

Notes for review

1. **TC_01 carries 19 valid input classes.** That is what the lecture asks for, but if it fails, the cause is unclear. If TC_01 fails at execution, split it before reporting the bug.
2. **TC_04 combines two valid classes** (latest OTP, password equal to current). If it fails, rerun each part on its own.
3. **TC_06 depends on luck:** the OTP must happen to start with `0`. If it hasn't after 30 tries, mark it "Not executed" with that reason.
4. **TC_20 assumes T ≤ 15 minutes.** If the OTP still works after 15 minutes, that is still a finding: SEC-07 says the OTP must expire.
5. **TC_07, TC_40, TC_41 send a `confirmPassword` field** whose name is assumed: the API spec has no such field.
6. **Expected error wording** is not in the spec; "required-field error", "incorrect-OTP error" and so on mean "a message with that meaning".